# Mobile Network Performance Analysis

Predicting Downlink PRB Utilization in 5G/LTE Radio Access Networks

Data Science Portfolio Project • 2025

Python • scikit-learn • pandas • seaborn

| 28,440 | 200 | 51 | 0.75 |
|---|---|---|---|
| Hourly Records | Network Sites | KPI Features | Final $R^2$ |

# Executive Summary

This project applies a complete data science workflow to a synthetic but realistic Radio Access Network (RAN) dataset — the infrastructure layer responsible for connecting mobile devices to the network. The central goal is to predict **Downlink Physical Resource Block (PRB) Utilization**, the industry-standard metric for cell-level capacity management in both LTE and 5G NR deployments.

Physical Resource Blocks are the fundamental unit of radio spectrum allocation. When PRB utilization climbs above ~80%, a cell is considered congested — users experience degraded throughput, increased latency, and higher drop rates. Accurately predicting utilization ahead of time allows network engineers to proactively redistribute load, schedule maintenance windows, and plan capacity expansions.

## Business Value

A trained PRB prediction model can function as a **Digital Twin** for capacity planners: simulate the impact of adding 5G subscribers, activating load balancing, or hosting a major event near a cell site — all before making changes to the live network. This reduces risk, lowers operational cost, and improves end-user quality of experience.

## Results Summary

| Stage | Approach | Outcome |
|---|---|---|
| Data Quality | Programmatic checks for nulls, negatives, out-of-range values | 100% clean — no imputation needed |
| EDA | Hourly trends, boxplots, correlation heatmaps | Peak-hour cycles identified; NR delivers ~3x LTE throughput |
| Baseline Model | Linear Regression | $R^2 \approx 0.46$ (non-linear patterns unresolved) |
| Improved Model | Random Forest + hour as categorical feature | $R^2 \approx 0.70$ (peak/off-peak captured) |
| Final Model | HistGradientBoosting + feature engineering | **$R^2 \approx 0.75$ RMSE reduced by ~35% vs baseline** |

# 1. Data Exploration & KPI Engineering

## Dataset Overview

The dataset contains **28,440 hourly records** generated across 200 simulated base station sites over multiple days. Each record represents one cell-hour and includes 51 columns spanning traffic counters, radio quality metrics, signalling KPIs, and operational flags.

| Dimension | Detail |
|---|---|
| Records | 28,440 hourly observations |
| Sites | 200 (160 Macro, 40 Indoor) |
| Technologies | LTE (4G) and NR (5G New Radio) |
| Frequency Bands | Multiple bands per technology |
| Key KPI Groups | Traffic, Signal Quality, Accessibility, Retainability, Mobility |
| Operational Flags | cell_outage, massive_event, load_balancing_active |

## Data Quality Validation

All quality checks were implemented as reusable functions to mirror production-grade data pipeline practices. Results confirmed:

- Zero null or missing values across all 51 columns
- Zero negative values in all counter and utilization columns
- Zero out-of-range values in rate and ratio columns (all confirmed within [0, 1])
- datetime column correctly parsed and validated

## KPI Engineering

Two additional features were derived to enrich the dataset beyond raw counters:

| Feature | Formula | Rationale |
|---|---|---|
| ho_sr_total | (intra + inter + inter-RAT HO success) / (total HO attempts) | Single combined mobility KPI across all handover types |
| traffic_per_user | dl_traffic_volume_gb / avg_active_users | Normalizes load by user count; captures per-user network stress |

# 2. Exploratory Data Analysis — Key Findings

## 2.1 Hourly Demand Patterns

Plotting hourly averages of active users, PRB utilization, and drop rate reveals a consistent daily cycle: network load is low between midnight and 08:00, ramps steeply during the morning commute, sustains a plateau through business and evening hours (09:00–20:00), then drops sharply overnight. This pattern aligns with human activity rhythms and is consistent across both technologies.

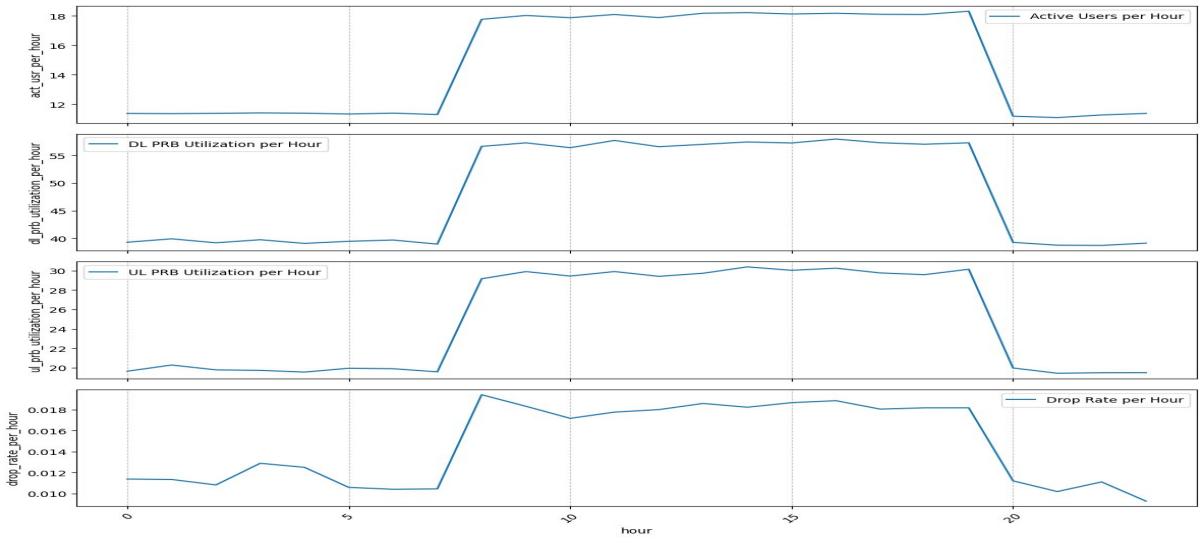


*Figure 1 — Hourly averages: active users, DL/UL PRB utilization, drop rate*

## 2.2 LTE vs 5G NR Performance

Boxplot comparisons confirm the expected capacity advantage of 5G NR. NR cells deliver a median DL throughput approximately **3 times higher** than LTE, with a much wider upper tail — reflecting the higher peak rates achievable on wider NR carriers. ERAB Setup Success Rate (accessibility) is statistically comparable between technologies, indicating that the core signalling plane performs consistently.

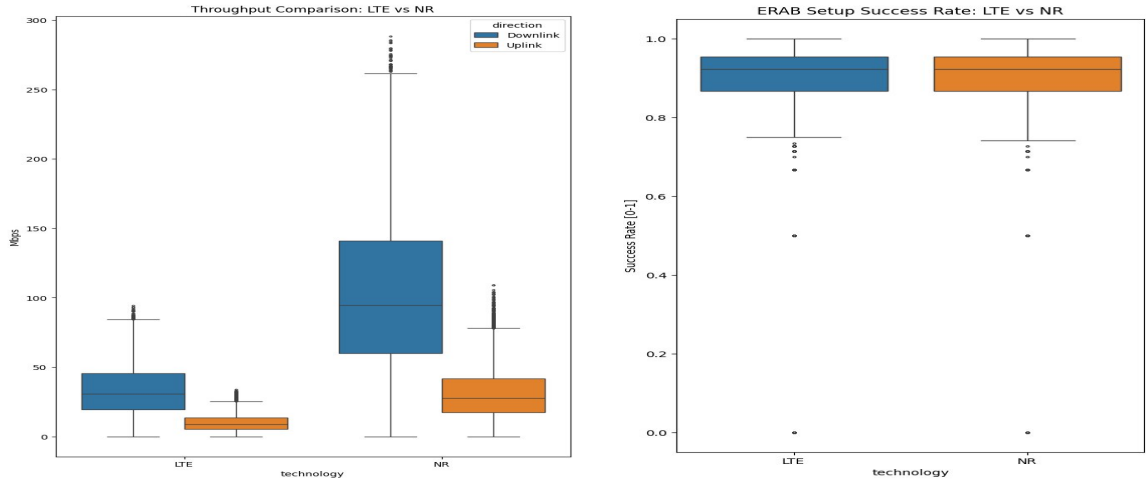


*Figure 2a — DL/UL Throughput by technology | Figure 2b — ERAB Setup Success Rate*

### 2.3 RSRP Distribution: Macro vs Indoor

Indoor cells show a median RSRP approximately **10 dBm better** than Macro (roughly -85 dBm vs -95 dBm), with tighter variance. This reflects the shorter propagation paths in indoor deployments. Macro cells show a longer lower tail extending below -120 dBm, characteristic of cell-edge users with poor coverage.

### 2.4 Correlation Analysis

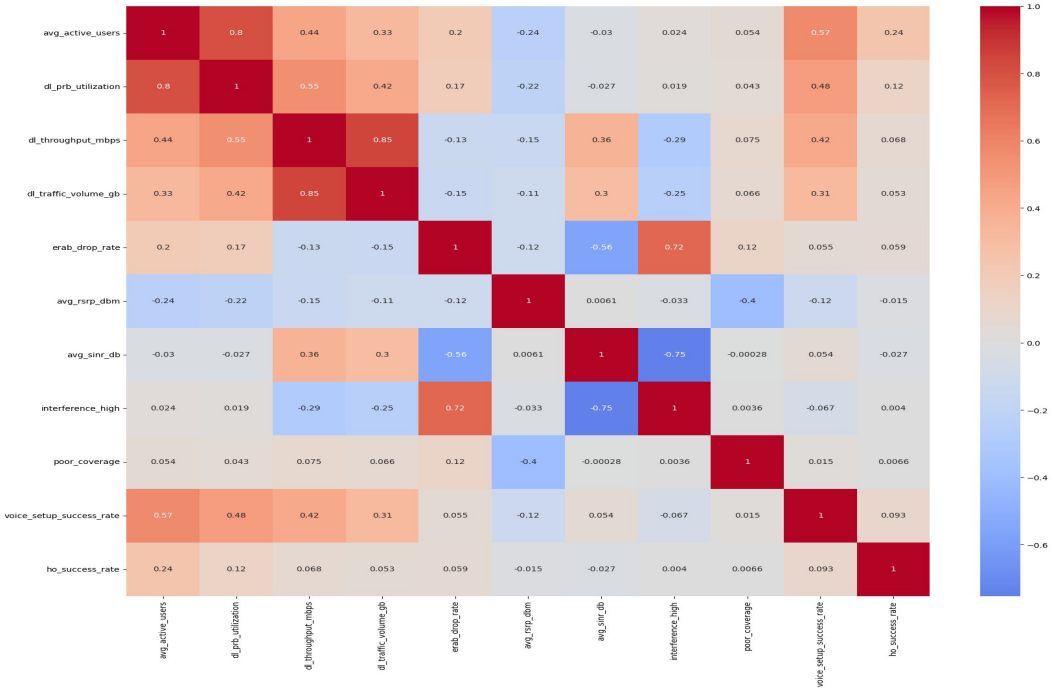


*Figure 3 — KPI subset correlation heatmap*

The heatmap identifies two well-defined correlation clusters:

- **Load & Traffic cluster:** avg_active_users, dl_prb_utilization, dl_throughput_mbps, dl_traffic_volume_gb are tightly correlated (r = 0.80–0.85), confirming that user demand is the primary driver of PRB consumption.
- **RF Quality cluster:** avg_sinr_db, interference_high, and erab_drop_rate are tightly linked (r = -0.75 and +0.72 respectively) — high interference degrades signal quality and directly increases call drop events.

## 3. Machine Learning — PRB Utilization Prediction

### 3.1 Encoding: pandas vs scikit-learn

The project explicitly compares two categorical encoding strategies to illustrate production best practices:

| | pd.get_dummies() | sklearn OneHotEncoder in Pipeline |
|---|---|---|
| **Data leakage risk** | Yes — test categories influence encoding | No — encoder fitted on train set only |
| **Unseen categories** | Error or NaN | Handled via handle_unknown='ignore' |
| **Pipeline integration** | Manual, error-prone | Native — single serialisable object |
| **Production deployment** | Not serialisable | joblib.dump() ready |

### 3.2 Model Iterations

Three successive model iterations were implemented, each diagnosing the previous model's weaknesses and addressing them systematically:

| Iteration | Model | Changes Made | $R^2$ |
|---|---|---|---|
| 1 | Linear Regression | Baseline; standard OHE features | ~0.46 |
| 2 | Random Forest | hour treated as categorical;<br>dl_traffic_volume_gb added | ~0.70 |
| **3 (Final)** | **HistGradientBoosting** | **Outage rows filtered;<br>traffic_per_user & traffic_sinr_ratio engineered;<br>massive_event & load_balancing_active included** | **~0.75** |

# 4. Conclusions & Next Steps

## Key Findings

- **Active users and traffic volume** are the strongest predictors of PRB utilization (correlation ~0.80), confirming that demand-side features should anchor any capacity model.
- **Situational flags** (massive_event, load_balancing_active, cell_outage) proved disproportionately impactful — their inclusion prevents systematic prediction errors during anomalous network states that would otherwise appear as model failures.
- **5G NR** delivers approximately 3x the median throughput of LTE at comparable PRB loads, validating the capacity case for NR rollout.
- **Indoor cells** show 10 dBm better median RSRP than Macro with lower variance — as expected for controlled RF propagation environments.
- **Interference management** is the primary lever for retainability: reducing interference_high directly reduces ERAB drop rate (r = 0.72).

## Model Limitations

The final model explains approximately 75% of variance in PRB utilization. The remaining ~25% reflects the inherent stochasticity of real radio networks: sudden user surges, momentary interference bursts, and device-level mobility that cannot be captured from hourly aggregates. Sub-hourly data (15-min counters, common in production OSS systems) would likely push $R^2$ above 0.85.

## Potential Extensions

| Extension | Expected Impact |
|---|---|
| 15-minute counter resolution | Reduce unexplained variance; capture intra-hour spikes |
| Anomaly detection layer | Flag cells deviating from predicted envelope in real time |
| Streamlit / Dash dashboard | Interactive KPI exploration by site, band, or technology |
| Model serving via REST API | Enable real-time inference on live network counter streams |
| Rolling-window retraining | Adapt to seasonal traffic shifts (holidays, events) |

*This report was generated as part of a data science portfolio project. The dataset is fully synthetic and intended for skill demonstration only. Full code and notebook available at: github.com/your-username/network-prb-prediction*